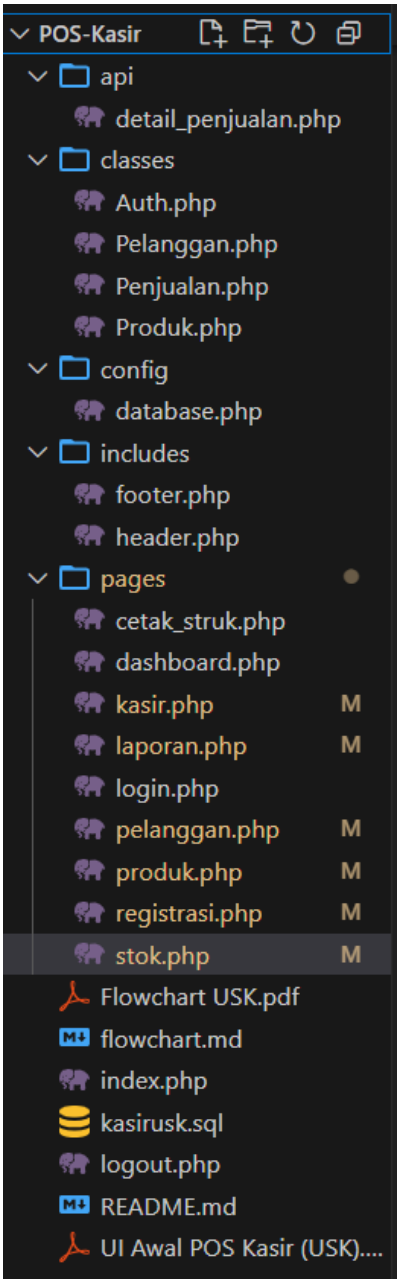
Dokumentasi USK Pembuatan Website POS System Kasir

A. Struktur Folder



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B.Flowchart

Text Editor Mermaid live

```
C.  graph TD
D.  A([Mulai]) --> B{Apakah sesi aktif?}
E.
F.      B -- Belum --> C[Halaman Login]
G.      C --> D{Cek Username & Password}
H.      D -- Tidak Valid --> C
I.      D -- Valid --> E{Cek Hak Akses}
J.
K.      B -- Sesi Aktif --> E
L.
M.      %% Cabang berdasarkan Hak Akses
N.      E -- Administrator --> F[Dashboard Admin]
O.      E -- Petugas/Kasir --> G[Halaman Transaksi Kasir]
P.
Q.      %% Alur Administrator
R.      F --> H[Kelola Data Produk CRUD]
S.      F --> I[Kelola Data Pengguna CRUD]
T.      F --> J[Lihat Laporan Penjualan]
U.      H --> P
V.      I --> P
W.      J --> P
X.
Y.      %% Alur Petugas/Kasir (Transaksi)
Z.      G --> K[Cari & Pilih Produk]
AA.     K --> L[Masukkan Produk ke Keranjang]
BB.     L --> M[Masukkan Nominal Pembayaran]
CC.     M --> N{Cek Pembayaran}
DD.     N --> O[Simpan Transaksi]
EE.     O --> P1[Cetak Struk/Nota]
FF.     P1 --> G
GG.
HH.     %% Logout
II.     P[Logout] --> Q([Selesai])
JJ.     G -. Jika ingin keluar .-> P
KK.     F -. Jika ingin keluar .-> P
LL.
```

C. Database

```
D.  -- phpMyAdmin SQL Dump
E.  -- version 5.2.0
F.  -- https://www.phpmyadmin.net/
G.  --
H.  -- Host: localhost:3306
I.  -- Generation Time: Apr 14, 2026 at 06:59 AM
J.  -- Server version: 8.0.30
K.  -- PHP Version: 8.1.10
L.
M.  SET SQL_MODE = "NO_AUTO_VALUE_ON_ZERO";
N.  START TRANSACTION;
O.  SET time_zone = "+00:00";
P.
Q.  /*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
R.  /*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
S.  /*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
T.  /*!40101 SET NAMES utf8mb4 */;
U.
V.  --
W.  -- Database: `kasirusk`
X.  --
Y.
Z.  -- --------------------------------------------------------
AA.
BB. --
CC. -- Table structure for table `detailpenjualan`
DD.--
EE.
FF. CREATE TABLE `detailpenjualan` (
GG.  `DetailID` int NOT NULL,
HH.  `PenjualanID` int NOT NULL,
II.   `ProdukID` int NOT NULL,
JJ.   `JumlahProduk` int NOT NULL DEFAULT '1',
KK.  `Subtotal` decimal(10,2) NOT NULL DEFAULT '0.00'
LL. ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
MM.
NN.--
OO.-- Dumping data for table `detailpenjualan`
PP. --
QQ.
```

```
RR. INSERT INTO `detailpenjualan` (`DetailID`, `PenjualanID`, `ProdukID`, `JumlahProduk`, `Subtotal`)
    VALUES
SS. (1, 1, 1, 2, '130000.00');
TT.
UU.-- -------------------------------------------------------
VV.
WW.         --
XX.-- Table structure for table `pelanggan`
YY.--
ZZ.
AAA.        CREATE TABLE `pelanggan` (
BBB.          `PelangganID` int NOT NULL,
CCC.          `NamaPelanggan` varchar(255) NOT NULL,
DDD.          `Alamat` text,
EEE.          `NomorTelepon` varchar(15) DEFAULT NULL
FFF.        ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
GGG.
HHH.        --
III.  -- Dumping data for table `pelanggan`
JJJ. --
KKK.
LLL.        INSERT INTO `pelanggan` (`PelangganID`, `NamaPelanggan`, `Alamat`, `NomorTelepon`)
    VALUES
MMM.        (1, 'Budi Santoso', 'Jl. Merdeka No.1 Samarinda', '081234567890'),
NNN.        (2, 'Siti Aminah', 'Jl. Pattimura No.5 Samarinda', '082345678901'),
OOO.        (3, 'Rahmat Hidayat', 'Jl. Sudirman No.10 Samarinda', '083456789012');
PPP.
QQQ.        -- -------------------------------------------------------
RRR.
SSS.        --
TTT.        -- Table structure for table `penjualan`
UUU.        --
VVV.
WWW.        CREATE TABLE `penjualan` (
XXX.          `PenjualanID` int NOT NULL,
YYY.          `TanggalPenjualan` date NOT NULL,
ZZZ.          `TotalHarga` decimal(10,2) NOT NULL DEFAULT '0.00',
AAAA.         `PelangganID` int DEFAULT NULL,
BBBB.         `UserID` int DEFAULT NULL
CCCC.       ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
DDDD.
EEEE.       --
FFFF.       -- Dumping data for table `penjualan`
GGGG.       --
HHHH.
```

```
IIII. INSERT INTO `penjualan` (`PenjualanID`, `TanggalPenjualan`, `TotalHarga`, `PelangganID`,
      `UserID`) VALUES
JJJJ.          (1, '2026-04-01', '130000.00', 1, 1);
KKKK.
LLLL.          -- -------------------------------------------------------
MMMM.
NNNN.          --
OOOO.          -- Table structure for table `produk`
PPPP.          --
QQQQ.
RRRR.          CREATE TABLE `produk` (
SSSS.            `ProdukID` int NOT NULL,
TTTT.            `NamaProduk` varchar(255) NOT NULL,
UUUU.            `Harga` decimal(10,2) NOT NULL DEFAULT '0.00',
VVVV.            `Stok` int NOT NULL DEFAULT '0'
WWWW.          ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
XXXX.
YYYY.          --
ZZZZ.          -- Dumping data for table `produk`
AAAAA.         --
BBBBB.
CCCCC.         INSERT INTO `produk` (`ProdukID`, `NamaProduk`, `Harga`, `Stok`) VALUES
DDDDD.         (1, 'Beras 5kg', '65000.00', 98),
EEEEE.         (2, 'Minyak Goreng 1L', '18000.00', 80),
FFFFF.         (3, 'Gula Pasir 1kg', '14000.00', 60),
GGGGG.         (4, 'Sabun Mandi', '5000.00', 150),
HHHHH.         (5, 'Shampo Sachet', '2000.00', 200);
IIIII.
JJJJJ.         -- -------------------------------------------------------
KKKKK.
LLLLL.         --
MMMMM.         -- Table structure for table `users`
NNNNN.         --
OOOOO.
PPPPP.         CREATE TABLE `users` (
QQQQQ.           `UserID` int NOT NULL,
RRRRR.           `NamaUser` varchar(100) NOT NULL,
SSSSS.           `Username` varchar(50) NOT NULL,
TTTTT.           `Password` varchar(255) NOT NULL,
UUUUU.           `Role` enum('administrator','petugas') NOT NULL DEFAULT 'petugas',
VVVVV.           `CreatedAt` timestamp NULL DEFAULT CURRENT_TIMESTAMP
WWWWW. ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
XXXXX.
YYYYY.         --
ZZZZZ.         -- Dumping data for table `users`
```

```
AAAAAA.  --
BBBBBB.
CCCCCC.   INSERT INTO `users` (`UserID`, `NamaUser`, `Username`, `Password`, `Role`, `CreatedAt`)
     VALUES
DDDDDD.  (1, 'Administrator', 'admin',
     '$2y$12$0XA66Vwm5YiSYMBoeKtzvOUF5RBKijbvVJiCgjpbwN3zqt/FfylgC', 'administrator',
     '2026-04-01 02:15:00'),
EEEEEE.    (2, 'Petugas Kasir', 'petugas',
     '$2y$12$ALSV9XEvOAeM2pC3Oj/pROAkWqGp9XNmjIKQP4CmoRmPiyvJ1s2BC', 'petugas',
     '2026-04-01 02:15:00');
FFFFFF.
GGGGGG.  --
HHHHHH.  -- Indexes for dumped tables
IIIIII.          --
JJJJJJ.
KKKKKK.  --
LLLLLL.    -- Indexes for table `detailpenjualan`
MMMMMM.          --
NNNNNN.  ALTER TABLE `detailpenjualan`
OOOOOO.    ADD PRIMARY KEY (`DetailID`),
PPPPPP.       ADD KEY `PenjualanID` (`PenjualanID`),
QQQQQQ.    ADD KEY `ProdukID` (`ProdukID`);
RRRRRR.
SSSSSS.     --
TTTTTT.    -- Indexes for table `pelanggan`
UUUUUU.  --
VVVVVV.  ALTER TABLE `pelanggan`
WWWWWW.           ADD PRIMARY KEY (`PelangganID`);
XXXXXX.
YYYYYY.  --
ZZZZZZ.    -- Indexes for table `penjualan`
AAAAAAA.          --
BBBBBBB. ALTER TABLE `penjualan`
CCCCCCC.   ADD PRIMARY KEY (`PenjualanID`),
DDDDDDD.             ADD KEY `PelangganID` (`PelangganID`),
EEEEEEE.    ADD KEY `UserID` (`UserID`);
FFFFFFF.
GGGGGGG.           --
HHHHHHH.           -- Indexes for table `produk`
IIIIIII.         --
JJJJJJJ.        ALTER TABLE `produk`
KKKKKKK.             ADD PRIMARY KEY (`ProdukID`);
LLLLLLL.
MMMMMMM.        --
NNNNNNN.           -- Indexes for table `users`
```

```
OOOOOOO.          --
PPPPPPP.   ALTER TABLE `users`
QQQQQQQ.            ADD PRIMARY KEY (`UserID`),
RRRRRRR.  ADD UNIQUE KEY `Username` (`Username`);
SSSSSSS.
TTTTTTT.  --
UUUUUUU.          -- AUTO_INCREMENT for dumped tables
VVVVVVV.          --
WWWWWWW.
XXXXXXX.          --
YYYYYYY.          -- AUTO_INCREMENT for table `detailpenjualan`
ZZZZZZZ.  --
AAAAAAAA.        ALTER TABLE `detailpenjualan`
BBBBBBBB.           MODIFY `DetailID` int NOT NULL AUTO_INCREMENT,
    AUTO_INCREMENT=2;
CCCCCCCC.
DDDDDDDD.        --
EEEEEEEE.-- AUTO_INCREMENT for table `pelanggan`
FFFFFFFF.  --
GGGGGGGG.        ALTER TABLE `pelanggan`
HHHHHHHH.          MODIFY `PelangganID` int NOT NULL AUTO_INCREMENT,
    AUTO_INCREMENT=4;
IIIIIIII.
JJJJJJJJ.      --
KKKKKKKK.        -- AUTO_INCREMENT for table `penjualan`
LLLLLLLL.--
MMMMMMMM.    ALTER TABLE `penjualan`
NNNNNNNN.          MODIFY `PenjualanID` int NOT NULL AUTO_INCREMENT,
    AUTO_INCREMENT=2;
OOOOOOOO.
PPPPPPPP.  --
QQQQQQQQ.        -- AUTO_INCREMENT for table `produk`
RRRRRRRR.          --
SSSSSSSS.  ALTER TABLE `produk`
TTTTTTTT.  MODIFY `ProdukID` int NOT NULL AUTO_INCREMENT, AUTO_INCREMENT=6;
UUUUUUUU.
VVVVVVVV.        --
WWWWWWWW.  -- AUTO_INCREMENT for table `users`
XXXXXXXX.        --
YYYYYYYY.        ALTER TABLE `users`
ZZZZZZZZ.  MODIFY `UserID` int NOT NULL AUTO_INCREMENT, AUTO_INCREMENT=3;
AAAAAAAAA.
BBBBBBBBB.       --
CCCCCCCCC.       -- Constraints for dumped tables
DDDDDDDDD.       --
```

```
EEEEEEEEE.
FFFFFFFFF.          --
GGGGGGGGG.      -- Constraints for table `detailpenjualan`
HHHHHHHHH.      --
IIIIIIIII.      ALTER TABLE `detailpenjualan`
JJJJJJJJJ.      ADD CONSTRAINT `detailpenjualan_ibfk_1` FOREIGN KEY (`PenjualanID`)
    REFERENCES `penjualan` (`PenjualanID`) ON DELETE CASCADE,
KKKKKKKKK.      ADD CONSTRAINT `detailpenjualan_ibfk_2` FOREIGN KEY (`ProdukID`)
    REFERENCES `produk` (`ProdukID`) ON DELETE RESTRICT;
LLLLLLLLL.
MMMMMMMMM. --
NNNNNNNNN.      -- Constraints for table `penjualan`
OOOOOOOOO.      --
PPPPPPPPP.      ALTER TABLE `penjualan`
QQQQQQQQQ.      ADD CONSTRAINT `penjualan_ibfk_1` FOREIGN KEY (`PelangganID`)
    REFERENCES `pelanggan` (`PelangganID`) ON DELETE SET NULL,
RRRRRRRRR.      ADD CONSTRAINT `penjualan_ibfk_2` FOREIGN KEY (`UserID`)
    REFERENCES `users` (`UserID`) ON DELETE SET NULL;
SSSSSSSSS.      COMMIT;
TTTTTTTTT.
UUUUUUUUU.      /*!40101 SET
    CHARACTER_SET_CLIENT=@OLD_CHARACTER_SET_CLIENT */;
VVVVVVVVV.      /*!40101 SET
    CHARACTER_SET_RESULTS=@OLD_CHARACTER_SET_RESULTS */;
WWWWWWWWW.      /*!40101 SET
    COLLATION_CONNECTION=@OLD_COLLATION_CONNECTION */;
XXXXXXXXX.
```

D. Kodingan

1. POS-Kasir/

a. Indeks.php

```
<?php
session_start();
if(isset($_SESSION['user_id'])) {
    header("Location: pages/dashboard.php");
    exit;
}
$error = '';
if($_SERVER['REQUEST_METHOD'] === 'POST') {
```

```
    require_once 'classes/Auth.php';
    $auth = new Auth();
    if($auth->login($_POST['username'], $_POST['password'])) {
        header("Location: pages/dashboard.php");
        exit;
    } else {
        $error = "Username atau password salah!";
    }
}
?>
<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Login – Aplikasi Kasir</title>
    <link
href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;600;700;800&display=swap"
rel="stylesheet">
    <script src="https://unpkg.com/@phosphor-icons/web"></script>
    <style>
    *, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }
    :root {
        --brand: #1a56db; --brand-dark: #1340a8; --accent: #f59e0b;
        --bg: #f0f4ff; --white: #ffffff; --text: #1e293b; --muted: #64748b;
        --danger: #ef4444; --radius: 14px;
    }
    body { font-family: 'Plus Jakarta Sans', sans-serif; background: var(--bg);
            display: flex; align-items: center; justify-content: center; min-height: 100vh; }
    .card {
        background: var(--white); border-radius: var(--radius); padding: 48px 40px;
        width: 100%; max-width: 420px; box-shadow: 0 20px 60px rgba(26,86,219,.15);
    }
    .logo { text-align: center; margin-bottom: 32px; }
    .logo-icon {
        width: 60px; height: 60px; background: var(--brand); border-radius: 16px;
        display: inline-flex; align-items: center; justify-content: center;
        font-size: 32px; margin-bottom: 12px; color: #fff;
    }
    h1 { font-size: 26px; font-weight: 800; color: var(--text); }
    p.sub { color: var(--muted); font-size: 14px; margin-top: 4px; }
    label { display: block; font-size: 13px; font-weight: 600; color: var(--text); margin-bottom: 6px; }
    input[type=text], input[type=password] {
        width: 100%; padding: 12px 16px; border: 2px solid #e2e8f0; border-radius: 10px;
        font-size: 15px; font-family: inherit; outline: none; transition: border .2s;
```

```
    }
    input:focus { border-color: var(--brand); }
    .field { margin-bottom: 18px; }
    .btn {
        width: 100%; padding: 14px; background: var(--brand); color: #fff;
        border: none; border-radius: 10px; font-size: 15px; font-weight: 700;
        cursor: pointer; transition: background .2s; margin-top: 8px;
    }
    .btn:hover { background: var(--brand-dark); }
    .error { background: #fef2f2; color: var(--danger); border-radius: 8px;
             padding: 10px 14px; font-size: 13px; margin-bottom: 16px; border: 1px solid #fecaca; }
    .hint { text-align:center; font-size:12px; color:var(--muted); margin-top:20px; }
    </style>
</head>
<body>
    <div class="card">
        <div class="logo">
            <div class="logo-icon"><i class="ph-fill ph-shopping-cart"></i></div>
            <h1>Aplikasi Kasir</h1>
            <p class="sub">Masuk untuk melanjutkan</p>
        </div>
        <?php if ($error): ?>
            <div class="error"><i class="ph-fill ph-warning-circle" style="vertical-align: text-bottom; font-size:
16px;"></i> <?= htmlspecialchars($error) ?></div>
        <?php endif; ?>
        <form method="POST">
            <div class="field">
                <label>Username</label>
                <input type="text" name="username" required autofocus autocomplete="username">
            </div>
            <div class="field">
                <label>Password</label>
                <input type="password" name="password" required autocomplete="current-password">
            </div>
            <button type="submit" class="btn">Masuk</button>
        </form>
    </div>
</body>
</html>
```

b. logout.php

```
<?php
require_once 'classes/Auth.php';
$auth = new Auth();
```

```
$auth->logout();
?>
```

2. api/

a. detail_penjualan.php

```
<?php
require_once '../config/database.php';
if (session_status() === PHP_SESSION_NONE) session_start();
if (!isset($_SESSION['user_id'])) { echo "Unauthorized"; exit; }

$db = Database::getInstance()->getConn();
$id = (int)($_GET['id'] ?? 0);
$stmt = $db->prepare(
    "SELECT d.*, pr.NamaProduk, pr.Harga FROM detailpenjualan d
     JOIN produk pr ON d.ProdukID = pr.ProdukID
     WHERE d.PenjualanID = ?"
);
$stmt->bind_param("i", $id);
$stmt->execute();
$rows = $stmt->get_result()->fetch_all(MYSQLI_ASSOC);
?>
<table style="width:100%;border-collapse:collapse;font-size:13px">
  <thead>
    <tr style="background:#f0f4ff">
      <th style="padding:8px 12px;text-align:left">Produk</th>
      <th style="padding:8px 12px">Harga Satuan</th>
      <th style="padding:8px 12px">Jumlah</th>
      <th style="padding:8px 12px">Subtotal</th>
    </tr>
  </thead>
  <tbody>
  <?php foreach ($rows as $r): ?>
    <tr>
      <td style="padding:8px 12px"><?= htmlspecialchars($r['NamaProduk']) ?></td>
      <td style="padding:8px 12px;text-align:center">Rp. <?= number_format($r['Harga'],0,',','.') ?></td>
      <td style="padding:8px 12px;text-align:center"><?= $r['JumlahProduk'] ?></td>
      <td style="padding:8px 12px;text-align:right"><strong>Rp. <?= number_format($r['Subtotal'],0,',','.')
?></strong></td>
    </tr>
  <?php endforeach; ?>
  </tbody>
</table>
```

3. classes/

a. Auth.php

```php
<?php
require_once __DIR__ . '/../config/database.php';

class Auth {
    private $conn;

    public function __construct() {
        $this->conn = Database::getInstance()->getConn();
    }

    public function login(string $username, string $password): bool {
        if (session_status() === PHP_SESSION_NONE) session_start();

        $stmt = $this->conn->prepare("SELECT * FROM users WHERE Username = ? LIMIT 1");
        $stmt->bind_param("s", $username);
        $stmt->execute();
        $result = $stmt->get_result();

        if ($row = $result->fetch_assoc()) {
            $valid = false;
            if (password_verify($password, $row['Password'])) {
                $valid = true;
            } elseif ($password === $row['Password']) {
                $newHash = password_hash($password, PASSWORD_DEFAULT);
                $upd = $this->conn->prepare("UPDATE users SET Password = ? WHERE UserID = ?");
                $upd->bind_param("si", $newHash, $row['UserID']);
                $upd->execute();
                $valid = true;
            }

            if ($valid) {
                $_SESSION['user_id']   = $row['UserID'];
                $_SESSION['username']  = $row['Username'];
                $_SESSION['nama']      = $row['NamaUser'];
                $_SESSION['role']      = $row['Role'];
                return true;
            }
        }
        return false;
    }
```

```
    public function logout(): void {
        if (session_status() === PHP_SESSION_NONE) session_start();
        session_destroy();
        header("Location: index.php");
        exit;
    }

    public static function cekLogin(): void {
        if (session_status() === PHP_SESSION_NONE) session_start();
        if (!isset($_SESSION['user_id'])) {
            header("Location: ../index.php");
            exit;
        }
    }

    public static function cekAdmin(): void {
        self::cekLogin();
        if ($_SESSION['role'] !== 'administrator') {
            header("Location: dashboard.php?error=akses_ditolak");
            exit;
        }
    }

    public function register(string $nama, string $username, string $password, string $role): bool {
        $hashed = password_hash($password, PASSWORD_DEFAULT);
        $stmt = $this->conn->prepare(
            "INSERT INTO users (NamaUser, Username, Password, Role) VALUES (?, ?, ?, ?)"
        );
        $stmt->bind_param("ssss", $nama, $username, $hashed, $role);
        return $stmt->execute();
    }

    public function getAll(): array {
        return $this->conn->query("SELECT * FROM users ORDER BY Role, NamaUser")-
>fetch_all(MYSQLI_ASSOC);
    }

    public function getById(int $id): ?array {
        $stmt = $this->conn->prepare("SELECT * FROM users WHERE UserID = ?");
        $stmt->bind_param("i", $id);
        $stmt->execute();
        $res = $stmt->get_result();
        return $res->fetch_assoc() ?: null;
    }
```

```
    public function edit(int $id, string $nama, string $username, string $password, string $role): bool {
        if ($password === '') {
            $stmt = $this->conn->prepare("UPDATE users SET NamaUser=?, Username=?, Role=? WHERE
UserID=?");
            $stmt->bind_param("sssi", $nama, $username, $role, $id);
        } else {
            $hashed = password_hash($password, PASSWORD_DEFAULT);
            $stmt = $this->conn->prepare("UPDATE users SET NamaUser=?, Username=?, Password=?, Role=?
WHERE UserID=?");
            $stmt->bind_param("ssssi", $nama, $username, $hashed, $role, $id);
        }
        return $stmt->execute();
    }

    public function hapus(int $id): bool {
        if ($_SESSION['user_id'] == $id) return false;
        $stmt = $this->conn->prepare("DELETE FROM users WHERE UserID = ?");
        $stmt->bind_param("i", $id);
        return $stmt->execute();
    }
}
```

b. Pelanggan.php

```
<?php
require_once __DIR__ . '/../config/database.php';

class Pelanggan {
    private $conn;

    public function __construct() {
        $this->conn = Database::getInstance()->getConn();
    }

    public function getAll(): array {
        return $this->conn->query("SELECT * FROM pelanggan ORDER BY NamaPelanggan")
                    ->fetch_all(MYSQLI_ASSOC);
    }

    public function getById(int $id): ?array {
        $stmt = $this->conn->prepare("SELECT * FROM pelanggan WHERE PelangganID=?");
        $stmt->bind_param("i", $id);
        $stmt->execute();
        return $stmt->get_result()->fetch_assoc() ?: null;
    }
```

```
    public function tambah(string $nama, string $alamat, string $telepon): bool {
        $stmt = $this->conn->prepare(
            "INSERT INTO pelanggan (NamaPelanggan, Alamat, NomorTelepon) VALUES (?,?,?)"
        );
        $stmt->bind_param("sss", $nama, $alamat, $telepon);
        return $stmt->execute();
    }

    public function edit(int $id, string $nama, string $alamat, string $telepon): bool {
        $stmt = $this->conn->prepare(
            "UPDATE pelanggan SET NamaPelanggan=?, Alamat=?, NomorTelepon=? WHERE PelangganID=?"
        );
        $stmt->bind_param("sssi", $nama, $alamat, $telepon, $id);
        return $stmt->execute();
    }

    public function hapus(int $id): bool {
        $stmt = $this->conn->prepare("DELETE FROM pelanggan WHERE PelangganID=?");
        $stmt->bind_param("i", $id);
        return $stmt->execute();
    }
}
```

c. Penjualan.php

```
<?php
require_once __DIR__ . '/../config/database.php';
require_once __DIR__ . '/Produk.php';

class Penjualan {
    private $conn;
    private $produk;

    public function __construct() {
        $this->conn  = Database::getInstance()->getConn();
        $this->produk = new Produk();
    }

    public function getAll(): array {
        $sql = "SELECT p.*, pl.NamaPelanggan, u.NamaUser
                FROM penjualan p
                LEFT JOIN pelanggan pl ON p.PelangganID = pl.PelangganID
                LEFT JOIN users u ON p.UserID = u.UserID
```

```
            ORDER BY p.TanggalPenjualan DESC, p.PenjualanID DESC";
        return $this->conn->query($sql)->fetch_all(MYSQLI_ASSOC);
    }

    public function getDetail(int $penjualanId): array {
        $stmt = $this->conn->prepare(
            "SELECT d.*, pr.NamaProduk FROM detailpenjualan d
             JOIN produk pr ON d.ProdukID = pr.ProdukID
             WHERE d.PenjualanID = ?"
        );
        $stmt->bind_param("i", $penjualanId);
        $stmt->execute();
        return $stmt->get_result()->fetch_all(MYSQLI_ASSOC);
    }

    /**
     * $items = [['produk_id'=>1,'jumlah'=>2], ...]
     */
    public function buat(int $pelangganId, int $userId, array $items): bool|int {
        $this->conn->begin_transaction();
        try {
            $total = 0;
            $details = [];
            foreach ($items as $item) {
                $p = $this->produk->getById((int)$item['produk_id']);
                if (!$p || $p['Stok'] < $item['jumlah']) {
                    throw new Exception("Stok tidak mencukupi: " . ($p['NamaProduk'] ?? ''));
                }
                $subtotal  = $p['Harga'] * $item['jumlah'];
                $total    += $subtotal;
                $details[] = ['produk_id'=>$p['ProdukID'],'jumlah'=>$item['jumlah'],'subtotal'=>$subtotal];
            }

            $tanggal = date('Y-m-d');
            $stmt = $this->conn->prepare(
                "INSERT INTO penjualan (TanggalPenjualan, TotalHarga, PelangganID, UserID) VALUES
(?,?,?,?)"
            );
            $stmt->bind_param("sdii", $tanggal, $total, $pelangganId, $userId);
            $stmt->execute();
            $penjualanId = $this->conn->insert_id;

            foreach ($details as $d) {
                $stmt2 = $this->conn->prepare(
```

```
                "INSERT INTO detailpenjualan (PenjualanID, ProdukID, JumlahProduk, Subtotal) VALUES
(?,?,?,?)"
            );
            $stmt2->bind_param("iiid", $penjualanId, $d['produk_id'], $d['jumlah'], $d['subtotal']);
            $stmt2->execute();
            $this->produk->kurangiStok($d['produk_id'], $d['jumlah']);
        }

        $this->conn->commit();
        return $penjualanId;
    } catch (Exception $e) {
        $this->conn->rollback();
        return false;
    }
  }

  public function hapus(int $id): bool {
    $stmt = $this->conn->prepare("DELETE FROM penjualan WHERE PenjualanID=?");
    $stmt->bind_param("i", $id);
    return $stmt->execute();
  }
}
```

d. Produk.php

```
<?php
require_once __DIR__ . '/../config/database.php';

class Produk {
  private $conn;

  public function __construct() {
    $this->conn = Database::getInstance()->getConn();
  }

  public function getAll(): array {
    $result = $this->conn->query("SELECT * FROM produk ORDER BY NamaProduk");
    return $result->fetch_all(MYSQLI_ASSOC);
  }

  public function getById(int $id): ?array {
    $stmt = $this->conn->prepare("SELECT * FROM produk WHERE ProdukID = ?");
    $stmt->bind_param("i", $id);
    $stmt->execute();
```

```
        $row = $stmt->get_result()->fetch_assoc();
        return $row ?: null;
    }

    public function tambah(string $nama, float $harga, int $stok): bool {
        $stmt = $this->conn->prepare(
            "INSERT INTO produk (NamaProduk, Harga, Stok) VALUES (?, ?, ?)"
        );
        $stmt->bind_param("sdi", $nama, $harga, $stok);
        return $stmt->execute();
    }

    public function edit(int $id, string $nama, float $harga, int $stok): bool {
        $stmt = $this->conn->prepare(
            "UPDATE produk SET NamaProduk=?, Harga=?, Stok=? WHERE ProdukID=?"
        );
        $stmt->bind_param("sdii", $nama, $harga, $stok, $id);
        return $stmt->execute();
    }

    public function hapus(int $id): bool {
        $stmt = $this->conn->prepare("DELETE FROM produk WHERE ProdukID=?");
        $stmt->bind_param("i", $id);
        return $stmt->execute();
    }

    public function kurangiStok(int $id, int $jumlah): bool {
        $stmt = $this->conn->prepare(
            "UPDATE produk SET Stok = Stok - ? WHERE ProdukID = ? AND Stok >= ?"
        );
        $stmt->bind_param("iii", $jumlah, $id, $jumlah);
        return $stmt->execute() && $stmt->affected_rows > 0;
    }
}
```

4. config/

a. database.php

```
<?php
define('DB_HOST', 'localhost');
define('DB_USER', 'root');
define('DB_PASS', '');
define('DB_NAME', 'kasir');
```

```php
class Database {
    private static $instance = null;
    private $conn;

    private function __construct() {
        $this->conn = new mysqli(DB_HOST, DB_USER, DB_PASS, DB_NAME);
        if ($this->conn->connect_error) {
            die("Koneksi gagal: " . $this->conn->connect_error);
        }
        $this->conn->set_charset("utf8mb4");
    }

    public static function getInstance(): Database {
        if (!self::$instance) {
            self::$instance = new Database();
        }
        return self::$instance;
    }

    public function getConn(): mysqli {
        return $this->conn;
    }
}
?>
```

5. includes/

a. footer.php

```php
</main>
</body>
</html>
```

b. header.php

```php
<?php
if (session_status() === PHP_SESSION_NONE) session_start();
$role = $_SESSION['role'] ?? '';
$nama = $_SESSION['nama'] ?? '';
$page = basename($_SERVER['PHP_SELF'], '.php');
?>
<!DOCTYPE html>
<html lang="id">
```

```
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width,initial-scale=1">
<title>Kasir – <?= ucfirst($page) ?></title>
<link
href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;700;800&display=sw
ap" rel="stylesheet">
<script src="https://unpkg.com/@phosphor-icons/web"></script>
<style>
  *, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }
  :root {
    --brand: #1a56db; --brand-dark: #1340a8; --brand-light: #eff6ff;
    --accent: #f59e0b; --bg: #f8fafc; --white: #fff;
    --text: #1e293b; --muted: #64748b; --border: #e2e8f0;
    --danger: #ef4444; --success: #10b981; --warning: #f59e0b;
    --sidebar-w: 240px; --radius: 12px;
  }
  body { font-family: 'Plus Jakarta Sans', sans-serif; background: var(--bg);
         color: var(--text); display: flex; min-height: 100vh; }

  .sidebar {
    width: var(--sidebar-w); background: var(--brand-dark); color: #fff;
    display: flex; flex-direction: column; position: fixed; top: 0; left: 0;
    height: 100vh; z-index: 100;
  }
  .sidebar-logo {
    padding: 24px 20px; display: flex; align-items: center; gap: 10px;
    border-bottom: 1px solid rgba(255,255,255,.12);
  }
  .sidebar-logo span { font-size: 22px; }
  .sidebar-logo h2 { font-size: 16px; font-weight: 800; letter-spacing: -.3px; }
  .sidebar-logo small { font-size: 11px; opacity: .6; }
  nav { flex: 1; padding: 16px 12px; }
  .nav-label { font-size: 10px; font-weight: 700; letter-spacing: 1px;
               text-transform: uppercase; opacity: .5; padding: 10px 8px 4px; }
  nav a {
    display: flex; align-items: center; gap: 10px; padding: 10px 12px;
    border-radius: 8px; color: rgba(255,255,255,.75); text-decoration: none;
    font-size: 14px; font-weight: 500; transition: all .15s; margin-bottom: 2px;
  }
  nav a:hover, nav a.active { background: rgba(255,255,255,.15); color: #fff; }
  nav a .icon { font-size: 20px; width: 24px; text-align: center; }
  .sidebar-footer {
    padding: 16px 12px; border-top: 1px solid rgba(255,255,255,.12);
  }
```

```
.sidebar-user { display: flex; align-items: center; gap: 10px; margin-bottom: 12px; }
.avatar {
  width: 36px; height: 36px; background: var(--accent); border-radius: 50%;
  display: flex; align-items: center; justify-content: center;
  font-weight: 800; font-size: 14px; color: #fff; flex-shrink: 0;
}
.sidebar-user-info small { display: block; font-size: 11px; opacity:.55; }
.sidebar-user-info strong { font-size: 13px; }
.btn-logout {
  display: flex; align-items: center; gap: 8px; width: 100%; padding: 9px 12px;
  background: rgba(239,68,68,.2); color: #fca5a5; border: none; border-radius: 8px;
  font-family: inherit; font-size: 13px; font-weight: 600; cursor: pointer;
  transition: background .15s;
}
.btn-logout:hover { background: rgba(239,68,68,.35); color: #fff; }

.main { margin-left: var(--sidebar-w); flex: 1; display: flex; flex-direction: column; }
.topbar {
  background: var(--white); border-bottom: 1px solid var(--border);
  padding: 16px 28px; display: flex; align-items: center; justify-content: space-between;
}
.topbar h1 { font-size: 20px; font-weight: 800; }
.content { padding: 28px; flex: 1; }

.card { background: var(--white); border-radius: var(--radius); border: 1px solid var(--border); overflow:
hidden; }
.card-header { padding: 18px 24px; border-bottom: 1px solid var(--border);
               display: flex; align-items: center; justify-content: space-between; }
.card-header h2 { font-size: 15px; font-weight: 700; }
.card-body { padding: 24px; }
.stat-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(180px, 1fr)); gap: 16px; margin-
bottom: 24px; }
.stat-card {
  background: var(--white); border-radius: var(--radius); padding: 20px;
  border: 1px solid var(--border); display: flex; align-items: center; gap: 16px;
}
.stat-icon { width: 48px; height: 48px; border-radius: 12px; display: flex;
             align-items: center; justify-content: center; font-size: 22px; flex-shrink:0; }
.stat-icon.blue   { background: #eff6ff; }
.stat-icon.green  { background: #f0fdf4; }
.stat-icon.amber  { background: #fffbeb; }
.stat-icon.red    { background: #fef2f2; }
.stat-label { font-size: 12px; color: var(--muted); font-weight: 500; }
.stat-value { font-size: 22px; font-weight: 800; }
```

```
  table { width: 100%; border-collapse: collapse; font-size: 14px; }
  th { background: var(--bg); padding: 10px 16px; text-align: left; font-size: 12px;
      font-weight: 700; color: var(--muted); text-transform: uppercase; letter-spacing: .5px; }
  td { padding: 12px 16px; border-bottom: 1px solid var(--border); vertical-align: middle; }
  tr:last-child td { border-bottom: none; }
  tr:hover td { background: #fafbff; }

  .badge {
    display: inline-block; padding: 3px 10px; border-radius: 99px;
    font-size: 11px; font-weight: 700;
  }
  .badge-blue   { background: var(--brand-light); color: var(--brand); }
  .badge-green  { background: #f0fdf4; color: #059669; }
  .badge-amber  { background: #fffbeb; color: #d97706; }
  .badge-red    { background: #fef2f2; color: var(--danger); }

  .btn { display: inline-flex; align-items: center; gap: 6px; padding: 9px 18px;
         border-radius: 8px; font-family: inherit; font-size: 13px; font-weight: 600;
         cursor: pointer; border: none; text-decoration: none; transition: all .15s; }
  .btn-primary   { background: var(--brand); color: #fff; }
  .btn-primary:hover { background: var(--brand-dark); }
  .btn-success   { background: var(--success); color: #fff; }
  .btn-danger    { background: var(--danger); color: #fff; }
  .btn-secondary { background: var(--bg); color: var(--text); border: 1px solid var(--border); }
  .btn-sm { padding: 5px 12px; font-size: 12px; border-radius: 6px; }

  .form-group { margin-bottom: 16px; }
  .form-group label { display: block; font-size: 13px; font-weight: 600; margin-bottom: 6px; }
  .form-control {
    width: 100%; padding: 10px 14px; border: 2px solid var(--border); border-radius: 8px;
    font-family: inherit; font-size: 14px; outline: none; transition: border .2s;
  }
  .form-control:focus { border-color: var(--brand); }
  .form-row { display: grid; gap: 16px; }
  .form-row.cols-2 { grid-template-columns: 1fr 1fr; }
  .form-row.cols-3 { grid-template-columns: 1fr 1fr 1fr; }

  .alert { padding: 12px 16px; border-radius: 8px; font-size: 13px; margin-bottom: 16px; }
  .alert-success { background: #f0fdf4; color: #059669; border: 1px solid #bbf7d0; }
  .alert-danger  { background: #fef2f2; color: var(--danger); border: 1px solid #fecaca; }
</style>
</head>
<body>
<aside class="sidebar">
  <div class="sidebar-logo">
```

```
    <span style="font-size: 28px; color: var(--brand-light);"><i class="ph-fill ph-shopping-cart-
simple"></i></span>
    <div><h2>Aplikasi Kasir</h2></div>
  </div>
  <nav>
    <div class="nav-label">Menu Utama</div>
    <a href="dashboard.php" class="<?= $page===='dashboard'?'active':'' ?>">
      <span class="icon"><i class="ph-fill ph-squares-four"></i></span> Dashboard
    </a>
    <a href="kasir.php" class="<?= $page===='kasir'?'active':'' ?>">
      <span class="icon"><i class="ph-fill ph-receipt"></i></span> Transaksi Kasir
    </a>
    <a href="stok.php" class="<?= $page===='stok'?'active':'' ?>">
      <span class="icon"><i class="ph-fill ph-package"></i></span> Stok Barang
    </a>
    <a href="laporan.php" class="<?= $page===='laporan'?'active':'' ?>">
      <span class="icon"><i class="ph-fill ph-chart-line-up"></i></span> Laporan Penjualan
    </a>
    <?php if ($role === 'administrator'): ?>
    <div class="nav-label">Administrator</div>
    <a href="produk.php" class="<?= $page===='produk'?'active':'' ?>">
      <span class="icon"><i class="ph-fill ph-tag"></i></span> Pendataan Barang
    </a>
    <a href="pelanggan.php" class="<?= $page===='pelanggan'?'active':'' ?>">
      <span class="icon"><i class="ph-fill ph-users"></i></span> Data Pelanggan
    </a>
    <a href="registrasi.php" class="<?= $page===='registrasi'?'active':'' ?>">
      <span class="icon"><i class="ph-fill ph-user-plus"></i></span> Manajemen User
    </a>
    <?php endif; ?>
  </nav>
  <div class="sidebar-footer">
    <div class="sidebar-user">
      <div class="avatar"><?= strtoupper(substr($nama,0,1)) ?></div>
      <div class="sidebar-user-info">
        <strong><?= htmlspecialchars($nama) ?></strong>
        <small><?= ucfirst($role) ?></small>
      </div>
    </div>
    <form action="../logout.php" method="POST">
      <button class="btn-logout" type="submit"><i class="ph-bold ph-sign-out" style="font-size: 16px;"></i>
Keluar</button>
    </form>
  </div>
</aside>
```

```
<main class="main">
```

6. pages/

a. cetak_struk.php

```
<?php
require_once '../classes/Auth.php';
Auth::cekLogin();
require_once '../config/database.php';

$id = (int)($_GET['id'] ?? 0);
if (!$id) {
    die("ID Transaksi tidak valid.");
}

$db = Database::getInstance()->getConn();

$stmt = $db->prepare(
    "SELECT p.*, pl.NamaPelanggan, u.NamaUser
     FROM penjualan p
     LEFT JOIN pelanggan pl ON p.PelangganID = pl.PelangganID
     LEFT JOIN users u ON p.UserID = u.UserID
     WHERE p.PenjualanID = ?"
);
$stmt->bind_param("i", $id);
$stmt->execute();
$transaksi = $stmt->get_result()->fetch_assoc();

if (!$transaksi) {
    die("Transaksi tidak ditemukan.");
}

$stmt = $db->prepare(
    "SELECT d.*, pr.NamaProduk, pr.Harga FROM detailpenjualan d
     JOIN produk pr ON d.ProdukID = pr.ProdukID
     WHERE d.PenjualanID = ?"
);
$stmt->bind_param("i", $id);
$stmt->execute();
$items = $stmt->get_result()->fetch_all(MYSQLI_ASSOC);

?>
<!DOCTYPE html>
```

```html
<html lang="id">
<head>
    <meta charset="UTF-8">
    <title>Struk Penjualan #<?= $transaksi['PenjualanID'] ?></title>
    <style>
        body {
            font-family: 'Courier New', Courier, monospace;
            font-size: 14px;
            margin: 0;
            padding: 20px;
            background: #fff;
            color: #000;
        }
        .struk-container {
            width: 300px;
            margin: 0 auto;
            border: 1px dashed #000;
            padding: 15px;
        }
        .header {
            text-align: center;
            margin-bottom: 20px;
        }
        .header h2 {
            margin: 0 0 5px 0;
            font-size: 18px;
        }
        .header p {
            margin: 0;
            font-size: 12px;
        }
        .info {
            margin-bottom: 15px;
            font-size: 12px;
        }
        .info table {
            width: 100%;
        }
        .info table td {
            padding: 2px 0;
        }
        .items {
            width: 100%;
            border-collapse: collapse;
            font-size: 12px;
```

```
        margin-bottom: 15px;
    }
    .items th, .items td {
        border-bottom: 1px dashed #000;
        padding: 5px 0;
    }
    .items th {
        text-align: left;
    }
    .items .right {
        text-align: right;
    }
    .items .center {
        text-align: center;
    }
    .total-area {
        font-size: 14px;
        font-weight: bold;
        margin-bottom: 20px;
    }
    .total-area table {
        width: 100%;
    }
    .total-area table td {
        padding: 2px 0;
    }
    .footer {
        text-align: center;
        font-size: 12px;
        border-top: 1px dashed #000;
        padding-top: 10px;
    }
    @media print {
        body {
            padding: 0;
        }
        .struk-container {
            border: none;
            width: 100%;
        }
        @page {
            margin: 0;
        }
    }
</style>
```

```
</head>
<body>
  <div class="struk-container">
    <div class="header">
      <h2>POS APP</h2>
      <p>Struk Pembelian</p>
    </div>

    <div class="info">
      <table>
        <tr>
          <td>No. Transaksi</td>
          <td>: #<?= $transaksi['PenjualanID'] ?></td>
        </tr>
        <tr>
          <td>Tanggal</td>
          <td>: <?= date('d/m/Y ', strtotime($transaksi['TanggalPenjualan'])) ?></td>
        </tr>
        <tr>
          <td>Kasir</td>
          <td>: <?= htmlspecialchars($transaksi['NamaUser'] ?? '-') ?></td>
        </tr>
        <tr>
          <td>Pelanggan</td>
          <td>: <?= htmlspecialchars($transaksi['NamaPelanggan'] ?? 'Umum') ?></td>
        </tr>
      </table>
    </div>

    <table class="items">
      <thead>
        <tr>
          <th>Item</th>
          <th class="center">Qty</th>
          <th class="right">Harga</th>
          <th class="right">Subtotal</th>
        </tr>
      </thead>
      <tbody>
        <?php foreach ($items as $item): ?>
        <tr>
          <td><?= htmlspecialchars($item['NamaProduk']) ?></td>
          <td class="center"><?= $item['JumlahProduk'] ?></td>
          <td class="right"><?= number_format($item['Harga'], 0, ',', '.') ?></td>
          <td class="right"><?= number_format($item['Subtotal'], 0, ',', '.') ?></td>
```

```
            </tr>
            <?php endforeach; ?>
          </tbody>
        </table>

        <div class="total-area">
          <table>
            <tr>
              <td>Total</td>
              <td class="right">Rp <?= number_format($transaksi['TotalHarga'], 0, ',', '.') ?></td>
            </tr>
          </table>
        </div>

        <div class="footer">
          <p>Terima Kasih Atas Kunjungan Anda</p>
        </div>
      </div>
      <script>
        window.addEventListener('load', () => {
          window.print();
        });
      </script>
    </body>
    </html>
```

b. dashboard.php

```
<?php
require_once '../classes/Auth.php';
Auth::cekLogin();
require_once '../config/database.php';
$db = Database::getInstance()->getConn();

$totalProduk    = $db->query("SELECT COUNT(*) FROM produk")->fetch_row()[0];
$totalPelanggan = $db->query("SELECT COUNT(*) FROM pelanggan")->fetch_row()[0];
$totalPenjualan = $db->query("SELECT COUNT(*) FROM penjualan WHERE TanggalPenjualan =
CURDATE()")->fetch_row()[0];
$totalPendapatan= $db->query("SELECT COALESCE(SUM(TotalHarga),0) FROM penjualan WHERE
TanggalPenjualan = CURDATE()")->fetch_row()[0];
$stokMenipis    = $db->query("SELECT * FROM produk WHERE Stok < 10 ORDER BY Stok")-
>fetch_all(MYSQLI_ASSOC);
$transaksiTerbaru = $db->query(
    "SELECT p.*, pl.NamaPelanggan FROM penjualan p
```

```
    LEFT JOIN pelanggan pl ON p.PelangganID = pl.PelangganID
    ORDER BY p.PenjualanID DESC LIMIT 5"
)->fetch_all(MYSQLI_ASSOC);

include '../includes/header.php';
?>
<div class="topbar"><h1><i class="ph-bold ph-squares-four"></i> Dashboard</h1><span style="font-
size:13px;color:var(--muted)"><?= date('d F Y') ?></span></div>
<div class="content">

  <div class="stat-grid">
    <div class="stat-card">
      <div class="stat-icon blue"><i class="ph-fill ph-receipt"></i></div>
      <div><div class="stat-label">Transaksi Hari Ini</div><div class="stat-value"><?= $totalPenjualan
?></div></div>
    </div>
    <div class="stat-card">
      <div class="stat-icon green"><i class="ph-fill ph-currency-circle-dollar"></i></div>
      <div><div class="stat-label">Pendapatan Hari Ini</div><div class="stat-value">Rp <?=
number_format($totalPendapatan,0,',','.') ?></div></div>
    </div>
    <div class="stat-card">
      <div class="stat-icon amber"><i class="ph-fill ph-package"></i></div>
      <div><div class="stat-label">Total Produk</div><div class="stat-value"><?= $totalProduk
?></div></div>
    </div>
  </div>

  <div style="display:grid;grid-template-columns:1fr 1fr;gap:20px">
    <div class="card">
      <div class="card-header"><h2>Transaksi Terbaru</h2>
        <a href="laporan.php" class="btn btn-secondary btn-sm">Lihat Semua</a>
      </div>
      <table>
        <thead><tr><th>#</th><th>Pelanggan</th><th>Total</th></tr></thead>
        <tbody>
        <?php foreach ($transaksiTerbaru as $t): ?>
          <tr>
            <td>#<?= $t['PenjualanID'] ?></td>
            <td><?= htmlspecialchars($t['NamaPelanggan'] ?? 'Umum') ?></td>
            <td><strong>Rp <?= number_format($t['TotalHarga'],0,',','.') ?></strong></td>
          </tr>
        <?php endforeach; ?>
        </tbody>
      </table>
```

```
  </div>

  <div class="card">
    <div class="card-header"><h2><i class="ph-bold ph-warning"></i> Stok Menipis (&lt;10)</h2></div>
    <table>
      <thead><tr><th>Produk</th><th>Sisa Stok</th></tr></thead>
      <tbody>
      <?php if (empty($stokMenipis)): ?>
        <tr><td colspan="2" style="text-align:center;color:var(--muted);padding:24px">Semua stok aman <i
class="ph-bold ph-check-circle" style="color:var(--success);"></i></td></tr>
      <?php else: ?>
        <?php foreach ($stokMenipis as $s): ?>
          <tr>
            <td><?= htmlspecialchars($s['NamaProduk']) ?></td>
            <td><span class="badge badge-red"><?= $s['Stok'] ?></span></td>
          </tr>
        <?php endforeach; ?>
      <?php endif; ?>
      </tbody>
    </table>
  </div>
 </div>
</div>
<?php include '../includes/footer.php'; ?>
```

c. kasir.php

```
<?php
require  once '../classes/Auth.php';
Auth::cekLogin();
require_once '../classes/Produk.php';
require_once '../classes/Pelanggan.php';
require  once '../classes/Penjualan.php';

$produkObj    = new Produk();
$pelangganObj = new Pelanggan();
$penjualanObj = new Penjualan();

$produkList    = $produkObj->getAll();
$pelangganList = $pelangganObj->getAll();

$msg = ''; $msgType = 'success';

if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['proses'])) {
```

```
    $pelangganId = (int)$_POST['pelanggan_id'];
    $items = [];
    if (!empty($_POST['produk_id'])) {
        foreach ($_POST['produk_id'] as $idx => $pid) {
            if ($pid && $_POST['jumlah'][$idx] > 0) {
                $items[] = ['produk_id' => (int)$pid, 'jumlah' => (int)$_POST['jumlah'][$idx]];
            }
        }
    }
    if (empty($items)) {
        $msg = "Pilih minimal satu produk!"; $msgType = 'danger';
    } else {
        $result = $penjualanObj->buat($pelangganId, $_SESSION['user_id'], $items);
        if ($result) {
            header("Location: kasir.php?sukses=" . $result); exit;
        } else {
            $msg = "Gagal menyimpan transaksi. Periksa stok barang."; $msgType = 'danger';
        }
    }
}

$sukses = isset($_GET['sukses']) ? (int)$_GET['sukses'] : null;

include '../includes/header.php';
?>
<div class="topbar"><h1><i class="ph-bold ph-receipt"></i> Transaksi Kasir</h1><span style="font-
size:13px;color:var(--muted)"><?= date('d F Y') ?></span></div>
<div class="content">
<?php if ($sukses): ?>
  <div class="alert alert-success"><i class="ph-bold ph-check-circle"></i> Transaksi #<?= $sukses ?> berhasil
disimpan!</div>
<?php endif; ?>
<?php if ($msg): ?>
  <div class="alert alert-<?= $msgType ?>"><?= htmlspecialchars($msg) ?></div>
<?php endif; ?>

<div style="display:grid;grid-template-columns:1.4fr 1fr;gap:24px;align-items:start">
  <div class="card">
    <div class="card-header"><h2>Form Transaksi</h2></div>
    <div class="card-body">
    <form method="POST" id="formKasir">
      <div class="form-group">
        <label>Pelanggan</label>
        <select name="pelanggan_id" class="form-control">
          <option value="0">-- Pilih Pelanggan --</option>
```

```
        <?php foreach ($pelangganList as $p): ?>
          <option value="<?= $p['PelangganID'] ?>"><?= htmlspecialchars($p['NamaPelanggan']) ?></option>
        <?php endforeach; ?>
      </select>
    </div>

    <div id="item-container">
      <div class="item-row form-row cols-3" style="margin-bottom:10px;align-items:end">
        <div class="form-group" style="margin:0">
          <label>Produk</label>
          <select name="produk_id[]" class="form-control produk-select" onchange="updateHarga(this)">
            <option value="">-- Pilih --</option>
            <?php foreach ($produkList as $pr): ?>
              <option value="<?= $pr['ProdukID'] ?>"
                      data-harga="<?= $pr['Harga'] ?>"
                      data-stok="<?= $pr['Stok'] ?>">
                <?= htmlspecialchars($pr['NamaProduk']) ?> (Stok: <?= $pr['Stok'] ?>)
              </option>
            <?php endforeach; ?>
          </select>
        </div>
        <div class="form-group" style="margin:0">
          <label>Jumlah</label>
          <input type="number" name="jumlah[]" class="form-control jumlah-input" min="1" value="1"
onchange="hitungTotal()">
        </div>
        <div class="form-group" style="margin:0">
          <label>Subtotal</label>
          <input type="text" class="form-control subtotal-display" readonly placeholder="Rp 0"
style="background:#f8fafc">
        </div>
      </div>
    </div>

    <button type="button" onclick="tambahBaris()" class="btn btn-secondary btn-sm" style="margin-
bottom:20px">
      <i class="ph-bold ph-plus"></i> Tambah Produk
    </button>

    <div style="background:var(--bg);border-radius:8px;padding:16px;margin-bottom:16px">
      <div style="display:flex;justify-content:space-between;font-size:13px;color:var(--muted)">
        <span>Total Belanja</span>
        <strong id="grand-total" style="font-size:20px;color:var(--brand)">Rp 0</strong>
      </div>
    </div>
```

```
    <button type="submit" name="proses" class="btn btn-primary" style="width:100%">
      <i class="ph-bold ph-credit-card"></i> Proses Pembayaran
    </button>
  </form>
  </div>
 </div>

 <div class="card">
  <div class="card-header"><h2><i class="ph-bold ph-package"></i> Daftar Produk</h2></div>
  <table>
    <thead><tr><th>Nama</th><th>Harga</th><th>Stok</th></tr></thead>
    <tbody>
    <?php foreach ($produkList as $pr): ?>
      <tr>
        <td><?= htmlspecialchars($pr['NamaProduk']) ?></td>
        <td>Rp <?= number_format($pr['Harga'],0,',','.') ?></td>
        <td>
          <span class="badge <?= $pr['Stok']<10?'badge-red':'badge-green' ?>">
            <?= $pr['Stok'] ?>
          </span>
        </td>
      </tr>
    <?php endforeach; ?>
    </tbody>
  </table>
 </div>
</div>
</div>

<script>
const produkData = <?= json_encode(array_column($produkList, null, 'ProdukID')) ?>;

function updateHarga(sel) { hitungTotal(); }

function hitungTotal() {
  let total = 0;
  document.querySelectorAll('.item-row').forEach(row => {
    const sel = row.querySelector('.produk-select');
    const jml = parseInt(row.querySelector('.jumlah-input').value) || 0;
    const sub = row.querySelector('.subtotal-display');
    if (sel.value) {
      const harga = parseFloat(sel.selectedOptions[0].dataset.harga) || 0;
      const subtotal = harga * jml;
      sub.value = 'Rp ' + subtotal.toLocaleString('id-ID');
      total += subtotal;
```

```
    } else {
      sub.value = '';
    }
  });
  document.getElementById('grand-total').textContent = 'Rp ' + total.toLocaleString('id-ID');
}

function tambahBaris() {
  const container = document.getElementById('item-container');
  const first = container.querySelector('.item-row');
  const clone = first.cloneNode(true);
  clone.querySelectorAll('input').forEach(i => i.value = i.type==='number'?1:'');
  clone.querySelector('select').selectedIndex = 0;
  const del = document.createElement('button');
  del.type = 'button'; del.innerHTML = '<i class="ph-bold ph-x"></i>';
  del.className = 'btn btn-danger btn-sm';
  del.style.marginTop = '4px';
  del.onclick = () => { clone.remove(); hitungTotal(); };
  clone.appendChild(del);
  container.appendChild(clone);
}
</script>
<?php include '../includes/footer.php'; ?>
```

d. laporan.php

```
<?php
require_once '../classes/Auth.php';
Auth::cekLogin();
require_once '../config/database.php';
require_once '../classes/Penjualan.php';

$db = Database::getInstance()->getConn();
$penjualanObj = new Penjualan();

$dari  = $_GET['dari']  ?? date('Y-m-01');
$sampai= $_GET['sampai']?? date('Y-m-d');

$stmt = $db->prepare(
    "SELECT p.*, pl.NamaPelanggan, u.NamaUser
     FROM penjualan p
     LEFT JOIN pelanggan pl ON p.PelangganID = pl.PelangganID
     LEFT JOIN users u ON p.UserID = u.UserID
     WHERE p.TanggalPenjualan BETWEEN ? AND ?
```

```
    ORDER BY p.TanggalPenjualan DESC, p.PenjualanID DESC"
);
$stmt->bind_param("ss", $dari, $sampai);
$stmt->execute();
$list = $stmt->get_result()->fetch_all(MYSQLI_ASSOC);

$totalPendapatan = array_sum(array_column($list, 'TotalHarga'));

include '../includes/header.php';
?>
<div class="topbar"><h1><i class="ph-bold ph-chart-line-up"></i> Laporan Penjualan</h1><span
style="font-size:13px;color:var(--muted)"><?= date('d F Y') ?></span></div>
<div class="content">

  <div class="card" style="margin-bottom:20px">
    <div class="card-body" style="padding:16px 24px">
    <form method="GET" style="display:flex;gap:16px;align-items:end;flex-wrap:wrap">
      <div class="form-group" style="margin:0">
        <label>Dari Tanggal</label>
        <input type="date" name="dari" class="form-control" value="<?= $dari ?>">
      </div>
      <div class="form-group" style="margin:0">
        <label>Sampai Tanggal</label>
        <input type="date" name="sampai" class="form-control" value="<?= $sampai ?>">
      </div>
      <button class="btn btn-primary"><i class="ph-bold ph-magnifying-glass"></i> Filter</button>
    </form>
    </div>
  </div>

  <div class="stat-grid" style="margin-bottom:20px">
    <div class="stat-card">
      <div class="stat-icon blue"><i class="ph-fill ph-receipt"></i></div>
      <div><div class="stat-label">Total Transaksi</div><div class="stat-value"><?= count($list)
?></div></div>
    </div>
    <div class="stat-card">
      <div class="stat-icon green"><i class="ph-fill ph-currency-circle-dollar"></i></div>
      <div><div class="stat-label">Total Pendapatan</div>
        <div class="stat-value">Rp <?= number_format($totalPendapatan,0,',','.') ?></div>
      </div>
    </div>
    <div class="stat-card">
      <div class="stat-icon amber"><i class="ph-fill ph-calendar"></i></div>
      <div><div class="stat-label">Periode</div>
```

```
      <div style="font-size:13px;font-weight:700"><?= date('d/m/Y',strtotime($dari)) ?> – <?=
date('d/m/Y',strtotime($sampai)) ?></div>
    </div>
  </div>
</div>

<div class="card">
  <div class="card-header"><h2>Detail Transaksi</h2>
    <!-- <a href="#" onclick="window.print()" class="btn btn-secondary btn-sm"><i class="ph-bold ph-
printer"></i> Cetak</a> -->
  </div>
  <table>
    <thead>
      <tr>
        <th>#ID</th><th>Tanggal</th><th>Pelanggan</th>
        <th>Kasir</th><th>Total</th><th>Aksi</th>
      </tr>
    </thead>
    <tbody>
    <?php if (empty($list)): ?>
      <tr><td colspan="6" style="text-align:center;padding:32px;color:var(--muted)">
        Tidak ada data pada periode ini.
      </td></tr>
    <?php else: ?>
      <?php foreach ($list as $p): ?>
        <tr>
          <td><strong>#<?= $p['PenjualanID'] ?></strong></td>
          <td><?= date('d/m/Y', strtotime($p['TanggalPenjualan'])) ?></td>
          <td><?= htmlspecialchars($p['NamaPelanggan'] ?? 'Umum') ?></td>
          <td><?= htmlspecialchars($p['NamaUser'] ?? '-') ?></td>
          <td><strong>Rp <?= number_format($p['TotalHarga'],0,',','.') ?></strong></td>
          <td style="display:flex;gap:8px">
            <button onclick="lihatDetail(<?= $p['PenjualanID'] ?>)"
                    class="btn btn-secondary btn-sm"><i class="ph-bold ph-eye"></i> Detail</button>
            <a href="cetak_struk.php?id=<?= $p['PenjualanID'] ?>" target="_blank" class="btn btn-primary btn-
sm"><i class="ph-bold ph-printer"></i> Cetak Struk</a>
          </td>
        </tr>
        <tr id="detail-<?= $p['PenjualanID'] ?>" style="display:none">
          <td colspan="6" style="background:#f8fafc;padding:0">
            <div style="padding:12px 24px" id="detail-content-<?= $p['PenjualanID'] ?>">
              Memuat...
            </div>
          </td>
        </tr>
```

```
      <?php endforeach; ?>
     <?php endif; ?>
     </tbody>
    </table>
  </div>
</div>

<script>
function lihatDetail(id) {
  const row = document.getElementById('detail-' + id);
  if (row.style.display === 'none') {
    row.style.display = '';
    fetch('../api/detail_penjualan.php?id=' + id)
      .then(r => r.text())
      .then(html => { document.getElementById('detail-content-' + id).innerHTML = html; });
  } else {
    row.style.display = 'none';
  }
}
</script>
<?php include '../includes/footer.php'; ?>
```

e. login.php

```
<?php
header("location: ../index.php");
exit;
```

f. pelanggan.php

```
<?php
require_once '../classes/Auth.php';
Auth::cekAdmin();
require_once '../classes/Pelanggan.php';
$pelangganObj = new Pelanggan();

$msg = ''; $msgType = 'success';
$edit = null;

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $nama    = trim($_POST['nama']);
    $alamat  = trim($_POST['alamat']);
    $telepon = trim($_POST['telepon']);
```

```
    if (isset($_POST['simpan'])) {
        $ok = $pelangganObj->tambah($nama, $alamat, $telepon);
        $msg = $ok ? "Pelanggan berhasil ditambahkan!" : "Gagal menambahkan pelanggan.";
        $msgType = $ok ? 'success' : 'danger';
    } elseif (isset($_POST['update'])) {
        $ok = $pelangganObj->edit((int)$_POST['id'], $nama, $alamat, $telepon);
        $msg = $ok ? "Data pelanggan diperbarui!" : "Gagal memperbarui data.";
        $msgType = $ok ? 'success' : 'danger';
    }
}

if (isset($_GET['hapus'])) {
    $ok = $pelangganObj->hapus((int)$_GET['hapus']);
    $msg = $ok ? "Pelanggan dihapus." : "Gagal menghapus pelanggan.";
    $msgType = $ok ? 'success' : 'danger';
}

if (isset($_GET['edit'])) {
    $edit = $pelangganObj->getById((int)$_GET['edit']);
}

$list = $pelangganObj->getAll();
include '../includes/header.php';
?>
<div class="topbar"><h1><i class="ph-bold ph-users"></i> Data Pelanggan</h1><span style="font-
size:13px;color:var(--muted)"><?= date('d F Y') ?></span></div>
<div class="content">
<?php if ($msg): ?>
  <div class="alert alert-<?= $msgType ?>"><?= htmlspecialchars($msg) ?></div>
<?php endif; ?>

<div style="display:grid;grid-template-columns:1fr 2fr;gap:24px;align-items:start">
  <div class="card">
    <div class="card-header"><h2><?= $edit ? '<i class="ph-bold ph-pencil-simple"></i> Edit Pelanggan' : '<i
class="ph-bold ph-plus"></i> Tambah Pelanggan' ?></h2></div>
    <div class="card-body">
    <form method="POST">
      <?php if ($edit): ?><input type="hidden" name="id" value="<?= $edit['PelangganID'] ?>"><?php endif;
?>
      <div class="form-group">
        <label>Nama Pelanggan</label>
        <input type="text" name="nama" class="form-control" required
               value="<?= htmlspecialchars($edit['NamaPelanggan'] ?? '') ?>">
      </div>
```

```
    <div class="form-group">
      <label>Alamat</label>
      <textarea name="alamat" class="form-control" rows="3"
                style="resize:vertical"><?= htmlspecialchars($edit['Alamat'] ?? '') ?></textarea>
    </div>
    <div class="form-group">
      <label>Nomor Telepon</label>
      <input type="text" name="telepon" class="form-control"
             value="<?= htmlspecialchars($edit['NomorTelepon'] ?? '') ?>">
    </div>
    <?php if ($edit): ?>
      <button name="update" class="btn btn-primary" style="width:100%"><i class="ph-bold ph-floppy-
disk"></i> Update</button>
      <a href="pelanggan.php" class="btn btn-secondary" style="width:100%;margin-top:8px;text-
align:center;display:block">Batal</a>
    <?php else: ?>
      <button name="simpan" class="btn btn-primary" style="width:100%"><i class="ph-bold ph-plus"></i>
Tambah</button>
    <?php endif; ?>
  </form>
  </div>
 </div>

 <div class="card">
  <div class="card-header">
    <h2>Daftar Pelanggan</h2>
    <span style="font-size:13px;color:var(--muted)"><?= count($list) ?> pelanggan</span>
  </div>
  <table>
    <thead>
      <tr><th>#</th><th>Nama</th><th>Telepon</th><th>Alamat</th><th>Aksi</th></tr>
    </thead>
    <tbody>
    <?php if (empty($list)): ?>
      <tr><td colspan="5" style="text-align:center;padding:24px;color:var(--muted)">Belum ada data
pelanggan.</td></tr>
    <?php else: ?>
      <?php foreach ($list as $i => $p): ?>
        <tr>
          <td><?= $i + 1 ?></td>
          <td><strong><?= htmlspecialchars($p['NamaPelanggan']) ?></strong></td>
          <td><?= htmlspecialchars($p['NomorTelepon'] ?: '-') ?></td>
          <td style="max-width:180px;overflow:hidden;text-overflow:ellipsis;white-space:nowrap">
            <?= htmlspecialchars($p['Alamat'] ?: '-') ?>
          </td>
```

```
          <td>
            <a href="?edit=<?= $p['PelangganID'] ?>" class="btn btn-secondary btn-sm"><i class="ph-bold ph-
pencil-simple"></i></a>
            <a href="?hapus=<?= $p['PelangganID'] ?>" class="btn btn-danger btn-sm"
              onclick="return confirm('Hapus pelanggan ini?')"><i class="ph-bold ph-trash"></i></a>
          </td>
        </tr>
      <?php endforeach; ?>
    <?php endif; ?>
    </tbody>
   </table>
  </div>
</div>
</div>
<?php include '../includes/footer.php'; ?>
```

g. produk.php

```
<?php
require_once '../classes/Auth.php';
Auth::cekAdmin();
require_once '../classes/Produk.php';
$produkObj = new Produk();

$msg = ''; $msgType = 'success';
$edit = null;

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $nama  = trim($_POST['nama']);
    $harga = (float)$_POST['harga'];
    $stok  = (int)$_POST['stok'];

    if (isset($_POST['simpan'])) {
        $ok = $produkObj->tambah($nama, $harga, $stok);
        $msg = $ok ? "Produk berhasil ditambahkan!" : "Gagal menambahkan produk.";
        $msgType = $ok ? 'success' : 'danger';
    } elseif (isset($_POST['update'])) {
        $ok = $produkObj->edit((int)$_POST['id'], $nama, $harga, $stok);
        $msg = $ok ? "Produk berhasil diupdate!" : "Gagal mengupdate produk.";
        $msgType = $ok ? 'success' : 'danger';
    }
}

if (isset($_GET['hapus'])) {
```

```
    $ok = $produkObj->hapus((int)$_GET['hapus']);
    $msg = $ok ? "Produk dihapus." : "Gagal menghapus produk.";
    $msgType = $ok ? 'success' : 'danger';
}

if (isset($_GET['edit'])) {
    $edit = $produkObj->getById((int)$_GET['edit']);
}

$list = $produkObj->getAll();
include '../includes/header.php';
?>
<div class="topbar"><h1><i class="ph-bold ph-tag"></i> Pendataan Barang</h1><span style="font-
size:13px;color:var(--muted)"><?= date('d F Y') ?></span></div>
<div class="content">
<?php if ($msg): ?>
  <div class="alert alert-<?= $msgType ?>"><?= htmlspecialchars($msg) ?></div>
<?php endif; ?>

<div style="display:grid;grid-template-columns:1fr 2fr;gap:24px;align-items:start">
  <div class="card">
    <div class="card-header"><h2><?= $edit ? '<i class="ph-bold ph-pencil-simple"></i> Edit Produk' : '<i
class="ph-bold ph-plus"></i> Tambah Produk' ?></h2></div>
    <div class="card-body">
    <form method="POST">
      <?php if ($edit): ?><input type="hidden" name="id" value="<?= $edit['ProdukID'] ?>"><?php endif; ?>
      <div class="form-group">
        <label>Nama Produk</label>
        <input type="text" name="nama" class="form-control" required
               value="<?= htmlspecialchars($edit['NamaProduk'] ?? '') ?>">
      </div>
      <div class="form-group">
        <label>Harga (Rp)</label>
        <input type="number" name="harga" class="form-control" min="0" step="100" required
               value="<?= $edit['Harga'] ?? '' ?>">
      </div>
      <div class="form-group">
        <label>Stok</label>
        <input type="number" name="stok" class="form-control" min="0" required
               value="<?= $edit['Stok'] ?? '' ?>">
      </div>
      <?php if ($edit): ?>
        <button name="update" class="btn btn-primary" style="width:100%"><i class="ph-bold ph-floppy-
disk"></i> Update</button>
```

```
      <a href="produk.php" class="btn btn-secondary" style="width:100%;margin-top:8px;text-
align:center">Batal</a>
    <?php else: ?>
      <button name="simpan" class="btn btn-primary" style="width:100%"><i class="ph-bold ph-plus"></i>
Tambah</button>
    <?php endif; ?>
  </form>
  </div>
 </div>

 <div class="card">
  <div class="card-header"><h2>Daftar Produk</h2>
    <span style="font-size:13px;color:var(--muted)"><?= count($list) ?> produk</span>
  </div>
  <table>
    <thead><tr><th>#</th><th>Nama
Produk</th><th>Harga</th><th>Stok</th><th>Aksi</th></tr></thead>
    <tbody>
    <?php foreach ($list as $i => $p): ?>
      <tr>
        <td><?= $i+1 ?></td>
        <td><?= htmlspecialchars($p['NamaProduk']) ?></td>
        <td>Rp <?= number_format($p['Harga'],0,',','.') ?></td>
        <td><span class="badge <?= $p['Stok']<10?'badge-red':'badge-green' ?>"><?= $p['Stok']
?></span></td>
        <td>
          <a href="?edit=<?= $p['ProdukID'] ?>" class="btn btn-secondary btn-sm"><i class="ph-bold ph-
pencil-simple"></i></a>
          <a href="?hapus=<?= $p['ProdukID'] ?>" class="btn btn-danger btn-sm"
            onclick="return confirm('Hapus produk ini?')"><i class="ph-bold ph-trash"></i></a>
        </td>
      </tr>
    <?php endforeach; ?>
    </tbody>
  </table>
 </div>
</div>
</div>
<?php include '../includes/footer.php'; ?>
```

h. registrasi.php

```
<?php
require_once '../classes/Auth.php';
```

```
Auth::cekAdmin();

$auth = new Auth();
$msg = ''; $msgType = 'success';
$edit = null;

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $nama     = trim($_POST['nama']);
    $username = trim($_POST['username']);
    $password = $_POST['password'];
    $role     = $_POST['role'];

    if (isset($_POST['simpan'])) {
        if (strlen($password) < 6) {
            $msg = "Password minimal 6 karakter."; $msgType = 'danger';
        } else {
            $ok = $auth->register($nama, $username, $password, $role);
            $msg = $ok ? "User berhasil didaftarkan!" : "Gagal: username mungkin sudah digunakan.";
            $msgType = $ok ? 'success' : 'danger';
        }
    } elseif (isset($_POST['update'])) {
        $id = (int)$_POST['id'];
        $ok = $auth->edit($id, $nama, $username, $password, $role);
        $msg = $ok ? "Data user diperbarui!" : "Gagal memperbarui data user.";
        $msgType = $ok ? 'success' : 'danger';
    }
}

if (isset($_GET['hapus'])) {
    $ok = $auth->hapus((int)$_GET['hapus']);
    $msg = $ok ? "User dihapus." : "Gagal menghapus user (tidak bisa menghapus akun sendiri).";
    $msgType = $ok ? 'success' : 'danger';
}

if (isset($_GET['edit'])) {
    $edit = $auth->getById((int)$_GET['edit']);
}

$users = $auth->getAll();

include '../includes/header.php';
?>
<div class="topbar"><h1><i class="ph-bold ph-user-plus"></i> Manajemen User</h1><span style="font-
size:13px;color:var(--muted)"><?= date('d F Y') ?></span></div>
<div class="content">
```

```
<?php if ($msg): ?>
  <div class="alert alert-<?= $msgType ?>"><?= htmlspecialchars($msg) ?></div>
<?php endif; ?>

<div style="display:grid;grid-template-columns:1fr 2fr;gap:24px;align-items:start">
  <div class="card">
    <div class="card-header"><h2><?= $edit ? '<i class="ph-bold ph-pencil-simple"></i> Edit User' : '<i
class="ph-bold ph-plus"></i> Tambah User' ?></h2></div>
    <div class="card-body">
    <form method="POST">
      <?php if ($edit): ?><input type="hidden" name="id" value="<?= $edit['UserID'] ?>"><?php endif; ?>
      <div class="form-group">
        <label>Nama Lengkap</label>
        <input type="text" name="nama" class="form-control" required
              value="<?= htmlspecialchars($edit['NamaUser'] ?? '') ?>">
      </div>
      <div class="form-group">
        <label>Username</label>
        <input type="text" name="username" class="form-control" required
              value="<?= htmlspecialchars($edit['Username'] ?? '') ?>">
      </div>
      <div class="form-group">
        <label>Password <?= $edit ? '<small style="color:var(--muted);font-weight:normal;font-
size:11px">(Kosongkan jika tidak diubah)</small>' : '' ?></label>
        <input type="password" name="password" class="form-control" <?= $edit ? '' : 'required minlength="6"'
?>>
      </div>
      <div class="form-group">
        <label>Role</label>
        <select name="role" class="form-control">
          <option value="petugas" <?= ($edit && $edit['Role'] === 'petugas') ? 'selected' : '' ?>>Petugas</option>
          <option value="administrator" <?= ($edit && $edit['Role'] === 'administrator') ? 'selected' : ''
?>>Administrator</option>
        </select>
      </div>
      <?php if ($edit): ?>
        <button name="update" class="btn btn-primary" style="width:100%"><i class="ph-bold ph-floppy-
disk"></i> Update User</button>
        <a href="registrasi.php" class="btn btn-secondary" style="width:100%;margin-top:8px;text-
align:center;display:block">Batal</a>
      <?php else: ?>
        <button name="simpan" class="btn btn-primary" style="width:100%"><i class="ph-bold ph-plus"></i>
Tambah User</button>
      <?php endif; ?>
    </form>
```

```
  </div>
 </div>

 <div class="card">
  <div class="card-header"><h2>Daftar User</h2>
   <span style="font-size:13px;color:var(--muted)"><?= count($users) ?> user</span>
  </div>
  <table>
   <thead><tr><th>#</th><th>Nama</th><th>Username</th><th>Role</th><th>Aksi</th></tr></thead>
   <tbody>
   <?php foreach ($users as $i => $u): ?>
    <tr>
     <td><?= $i+1 ?></td>
     <td><?= htmlspecialchars($u['NamaUser']) ?></td>
     <td><code><?= htmlspecialchars($u['Username']) ?></code></td>
     <td><span class="badge <?= $u['Role']==='administrator'?'badge-blue':'badge-green' ?>">
      <?= ucfirst($u['Role']) ?>
     </span></td>
     <td>
      <a href="?edit=<?= $u['UserID'] ?>" class="btn btn-secondary btn-sm"><i class="ph-bold ph-pencil-
simple"></i></a>
      <?php if ($u['UserID'] != $_SESSION['user_id']): ?>
       <a href="?hapus=<?= $u['UserID'] ?>" class="btn btn-danger btn-sm"
        onclick="return confirm('Hapus user ini?')"><i class="ph-bold ph-trash"></i></a>
      <?php else: ?>
       <button disabled class="btn btn-secondary btn-sm" style="opacity:0.5;cursor:not-allowed"
title="Anda tidak bisa menghapus akun sendiri"><i class="ph-bold ph-trash"></i></button>
      <?php endif; ?>
     </td>
    </tr>
   <?php endforeach; ?>
   </tbody>
  </table>
 </div>
</div>
</div>
<?php include '../includes/footer.php'; ?>
```

i. stok.php

```
<?php
require_once '../classes/Auth.php';
Auth::cekLogin();
require_once '../classes/Produk.php';
```

```
$produkObj = new Produk();
$list = $produkObj->getAll();
include '../includes/header.php';
?>
<div class="topbar"><h1><i class="ph-bold ph-package"></i> Stok Barang</h1><span style="font-
size:13px;color:var(--muted)"><?= date('d F Y') ?></span></div>
<div class="content">
  <div class="card">
    <div class="card-header">
      <h2>Status Stok Semua Produk</h2>
      <span style="font-size:13px;color:var(--muted)"><?= count($list) ?> produk</span>
    </div>
    <table>
      <thead>
        <tr><th>#</th><th>Nama Produk</th><th>Harga</th><th>Stok</th><th>Status</th></tr>
      </thead>
      <tbody>
      <?php foreach ($list as $i => $p): ?>
        <?php
          $status = $p['Stok'] === 0 ? ['Habis','badge-red'] :
                  ($p['Stok'] < 10  ? ['Menipis','badge-amber'] : ['Tersedia','badge-green']);
        ?>
        <tr>
          <td><?= $i+1 ?></td>
          <td><?= htmlspecialchars($p['NamaProduk']) ?></td>
          <td>Rp <?= number_format($p['Harga'],0,',','.') ?></td>
          <td>
            <div style="display:flex;align-items:center;gap:8px">
              <div style="flex:1;background:#e2e8f0;border-radius:99px;height:8px;min-width:80px">
                <div style="background:<?= $p['Stok']<10?'#f59e0b':'#10b981' ?>;
                        width:<?= min(100, $p['Stok']) ?>%;height:8px;border-radius:99px"></div>
              </div>
              <strong><?= $p['Stok'] ?></strong>
            </div>
          </td>
          <td><span class="badge <?= $status[1] ?>"><?= $status[0] ?></span></td>
        </tr>
      <?php endforeach; ?>
      </tbody>
    </table>
  </div>
</div>
<?php include '../includes/footer.php'; ?>
```

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